

Winston Zhang

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Final year Mechanical Engineering student with a focus on simulation-driven iterative design. Proven hands-on experience in aerodynamics, CFD, FEA and composite manufacturing for high-performance applications. Led the design and validation process of FSAE aerodynamic systems and contributed to the drag-reduction optimization process in sustainable electric aviation.

EXPERIENCE

SYDNEY MOTORSPORTS FORMULA STUDENT TEAM

Sydney, Australia

CAD | Aerodynamic Design | CFD Validation | Composite Manufacturing

Advisor

February 2025 - present

Working with the aerodynamics team to validate the full aero package using the University's wind tunnel as part of graduate thesis research. Provide technical guidance on CFD validation and composite manufacturing based on prior leadership experience within the team.

Aerodynamics Section Lead

December 2023 - February 2025

Led a team of 10 student engineers in the in-house development and manufacturing of a full aerodynamics package using SOLIDWORKS, ANSYS Fluent, laser cutting, and 3D printing. Delivered a structurally robust aero package with improved rigidity over previous years while maintaining a strict 3 kg weight target. Designed a rib and spar system for the front wing to handle unexpected loads during testing and competition. Coordinated large assemblies (>1000 parts), enforced compliance with FSAE rules, and managed timelines and task delegation.

Aerodynamics Engineer

July 2022 - December 2023

Designed and validated the front wing and undertray of the SM23 vehicle using CFD. Conducted iterative design processes using a large computing cluster to optimize aerodynamic performance. Participated in mold design and manufacturing, and performed carbon fiber layups using prepreg and conventional wet-layup techniques to produce high-performance composite components.

DOVETAIL ELECTRIC AVIATION

Sydney, Australia

CAD | CFD Validation | Reverse Engineering

Computational Fluid Dynamics Engineer (Intern)

September 2023 - January 2024

Reverse-engineered a propeller nacelle assembly by generating parametric CAD models from 3D scans in SOLIDWORKS.. Validated scanned geometry in ANSYS Fluent, optimizing cooling duct geometries to minimize drag and improve airflow efficiency. Created efficient CFD models to reduce simulation time and resource consumption.

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics)

Expected Graduation: 2025

2021 - Current

TECHNICAL SKILLS

Software

- Created mechanical designs with DFM principles in mind in SOLIDWORKS and Autodesk Inventor
- Conducted FEA Analysis using SOLIDWORKS' built in simulation packages
- Conducted various numerical simulations such as FEA, topology studies, CFD and thermal analysis through the ANSYS Suite
- Wrote MATLAB scrips for simple numerical simulations such as 2D CFD, BEM as well as various data-parsing programs.
- Proficient in Python through various calculation and simulation scripts.
- Literate in the Microsoft Office Suite

Manufacturing

- Create composite parts with prepreg and conventional wet-layup techniques.
- Create composite molds with modern techniques such as hot wire cutting, laser cutting, 3D printing and conventional techniques such as hand-sculpting foam molds.
- 3D Printing and laser cutting for rapid manufacturing.
- Extensive experience in hand tools and power tools

TECHNICAL PROJECTS

- **Homelab Server** - Functions as both a local media server and a self-hosted cloud storage solution, enabling high-speed remote access and data redundancy.
- **Robotic Arm** - A 6-axis robotic arm powered by NEMA stepper motors, with motion transmitted via belt drives to reduce backlash. The accuracy was enhanced through a custom-developed zero-backlash cycloidal gearbox tailored for precise torque and speed control.
- **Custom PC Peripherals** - Engineered a 32-gram fingertip grip mouse optimized for competitive FPS gaming, and assembled a Lily58 split ortholinear keyboard with a custom-designed case for improved ergonomics and portability.
- **UNIX** - Daily-operated dual-boot Windows/Linux setups, leveraging distros such as Arch and Ubuntu derivatives for development flexibility, customization, and deep system-level understanding.